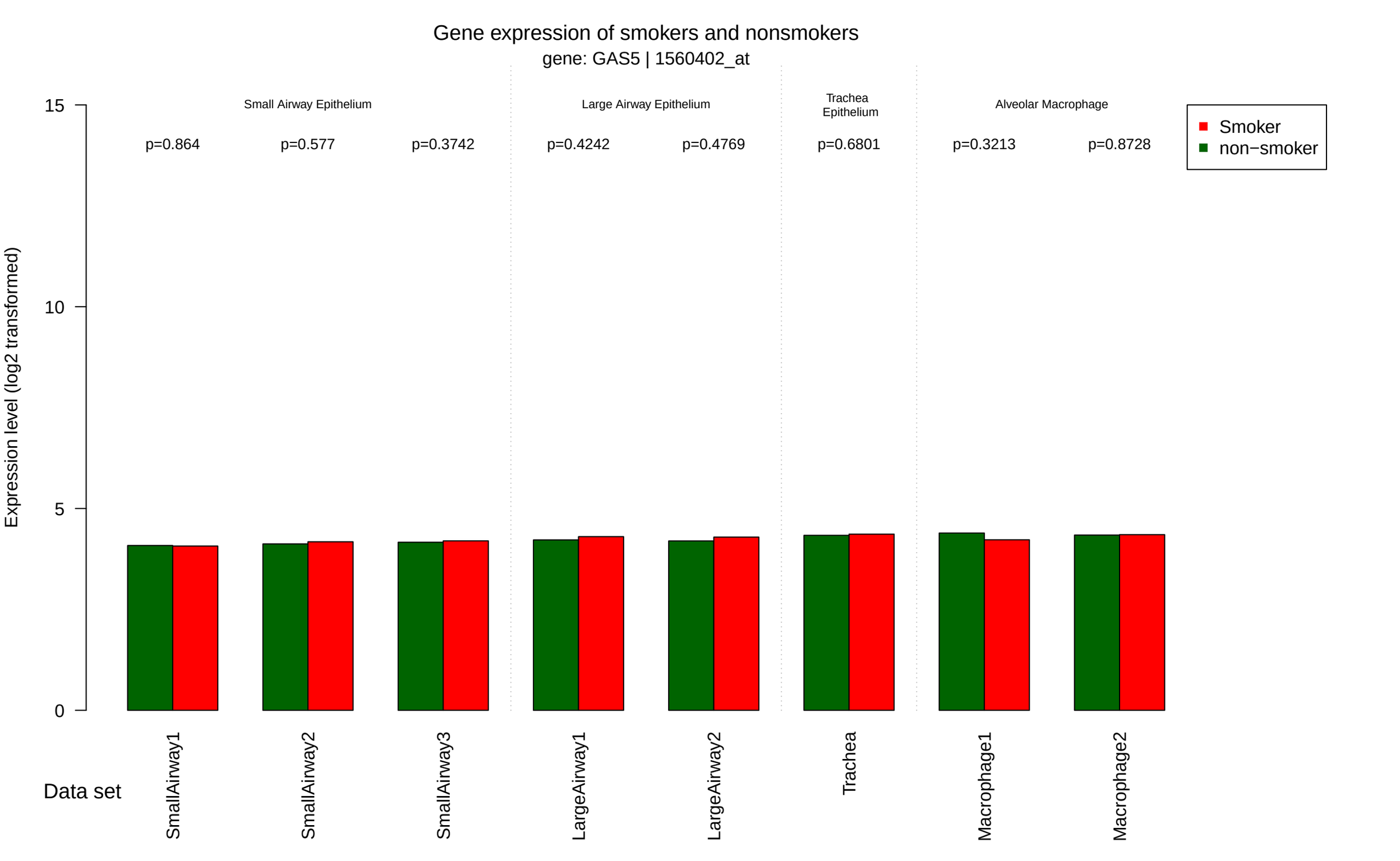
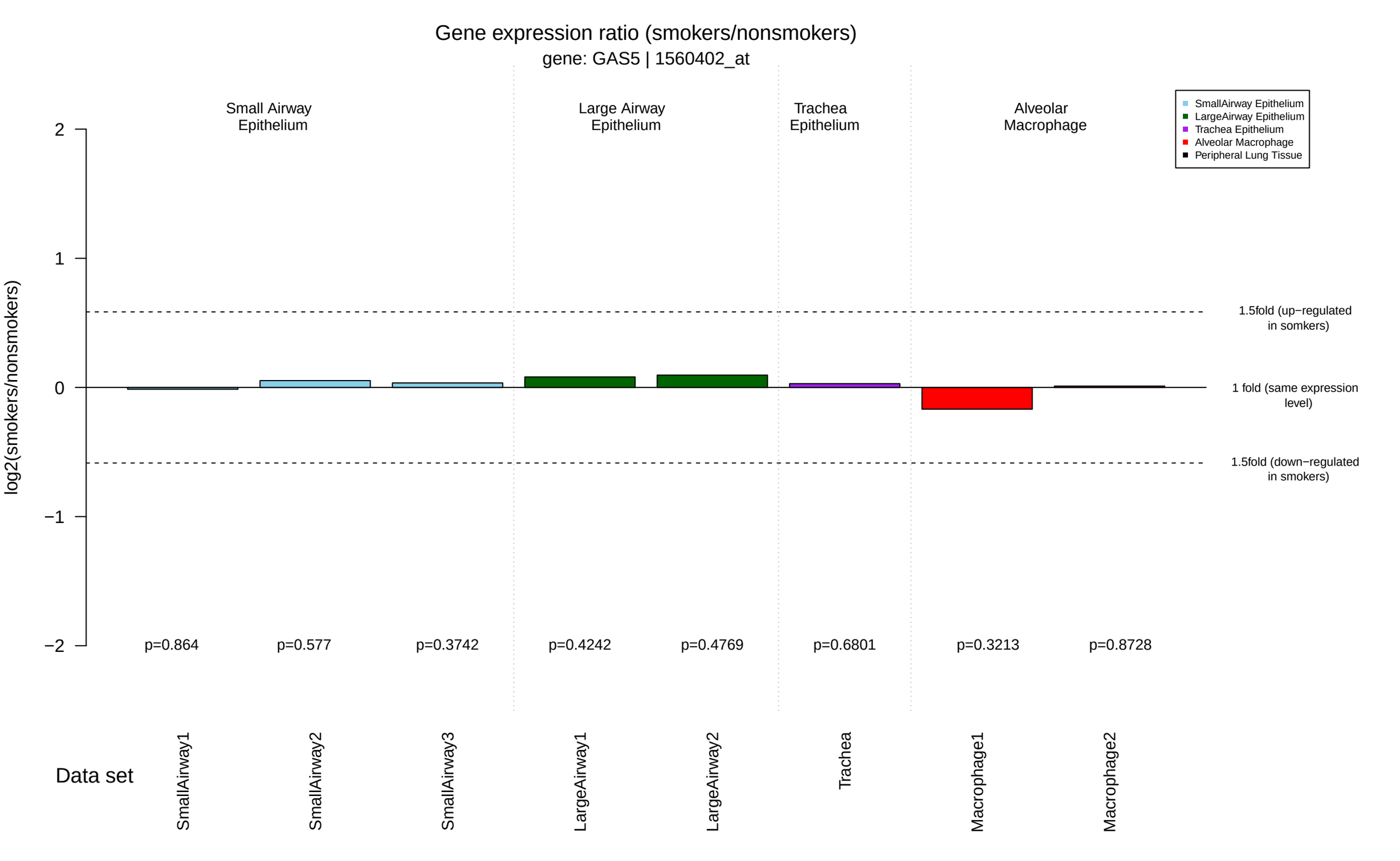
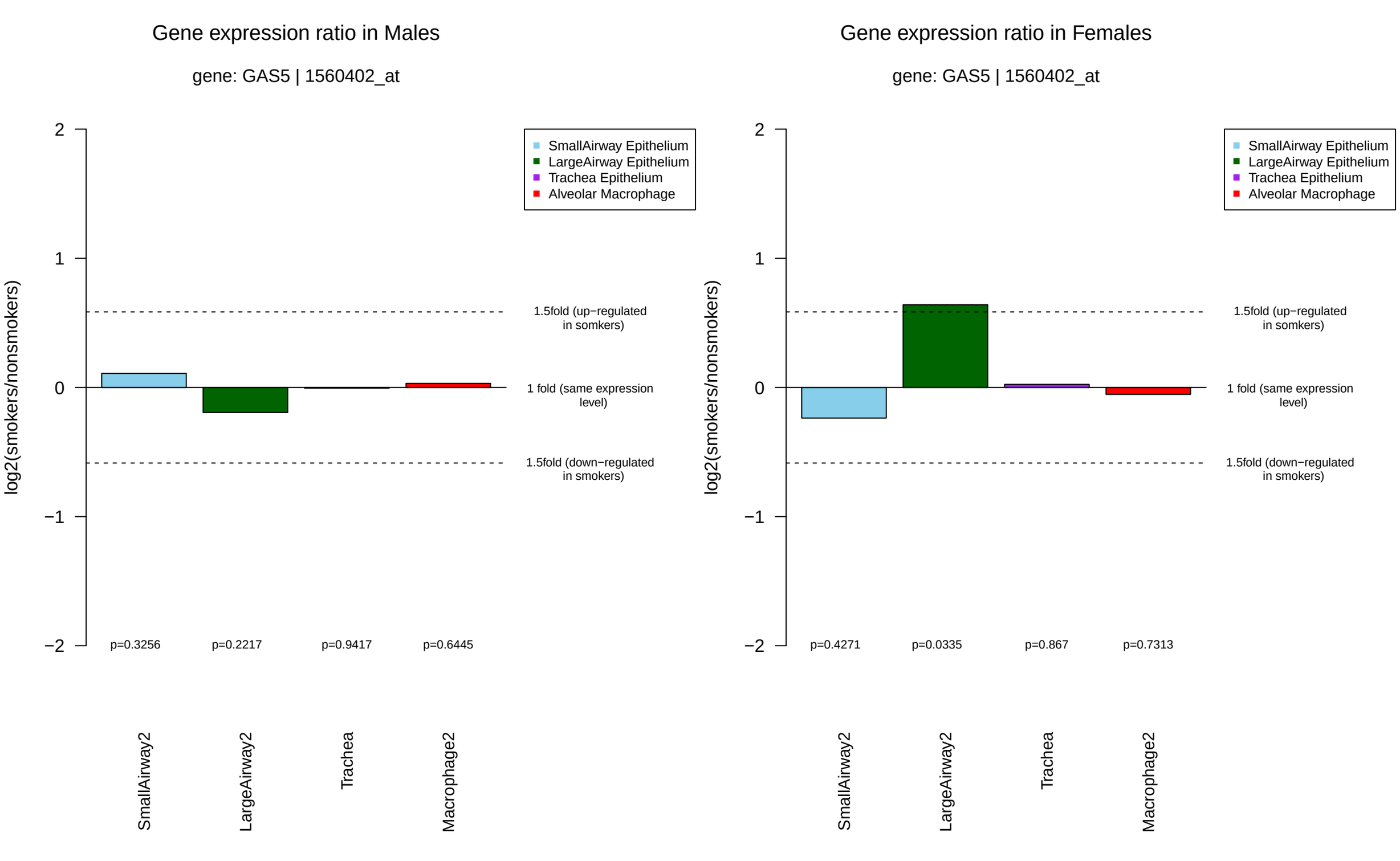
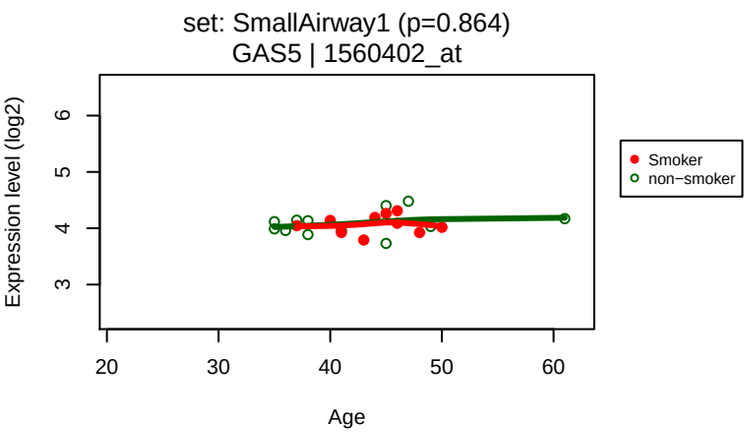
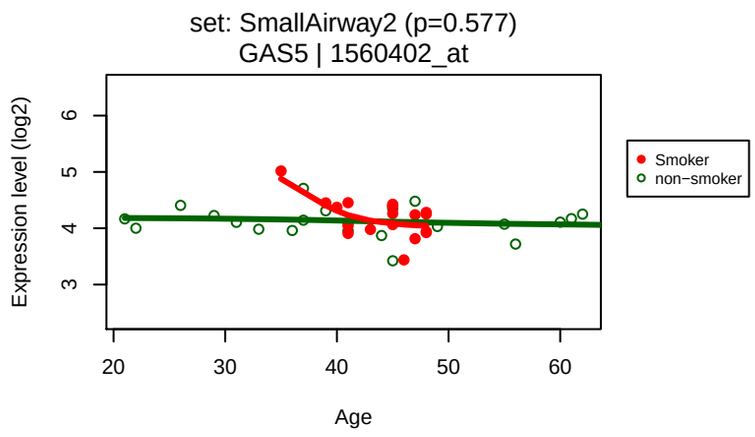
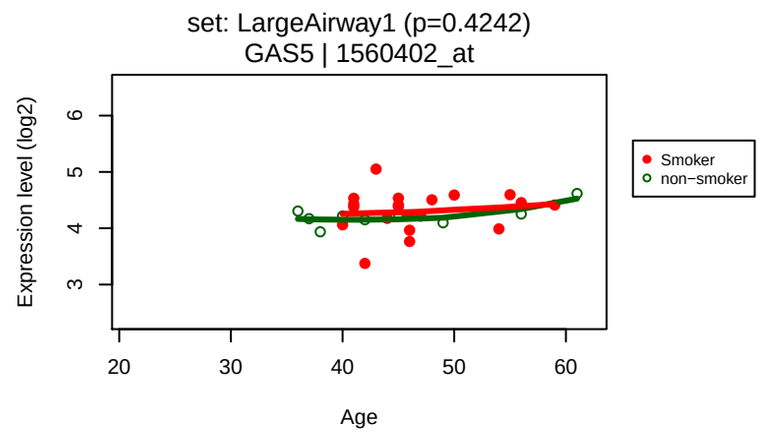

Gene expression of smokers and nonsmokers
gene: GAS5 | 1560402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.864
p=0.577
p=0.3742
p=0.4242
p=0.4769
p=0.6801
p=0.3213
p=0.8728
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

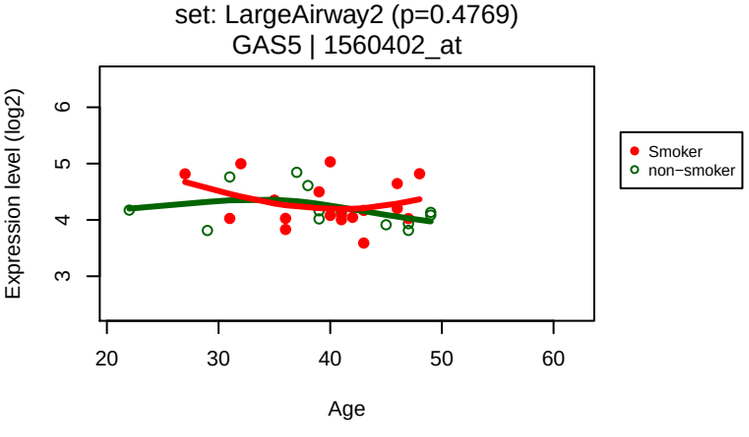

Gene expression ratio (smokers/nonsmokers)
gene: GAS5 | 1560402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.864
p=0.577
p=0.3742
p=0.4242
p=0.4769
p=0.6801
p=0.3213
p=0.8728
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

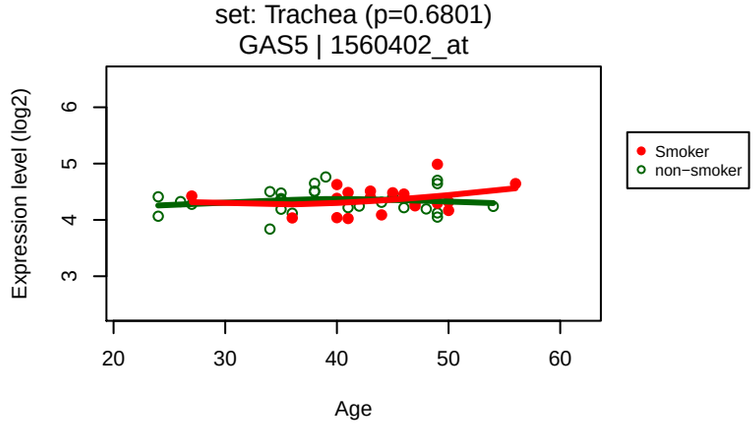

Gene expression ratio in Males
gene: GAS5 | 1560402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3256
p=0.2217
p=0.9417
p=0.6445
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAS5 | 1560402_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4271
p=0.0335
p=0.867
p=0.7313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

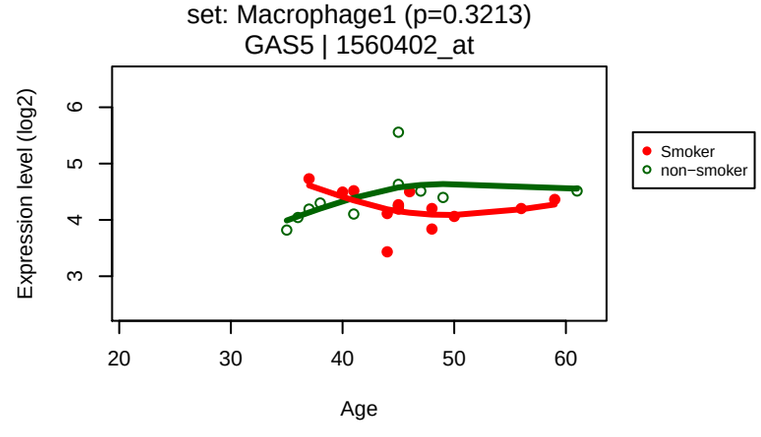

set: SmallAirway1 (p=0.864)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker

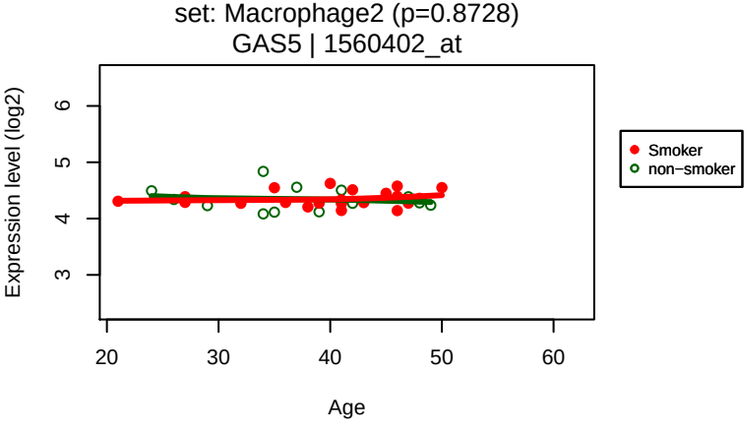

set: SmallAirway2 (p=0.577)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker

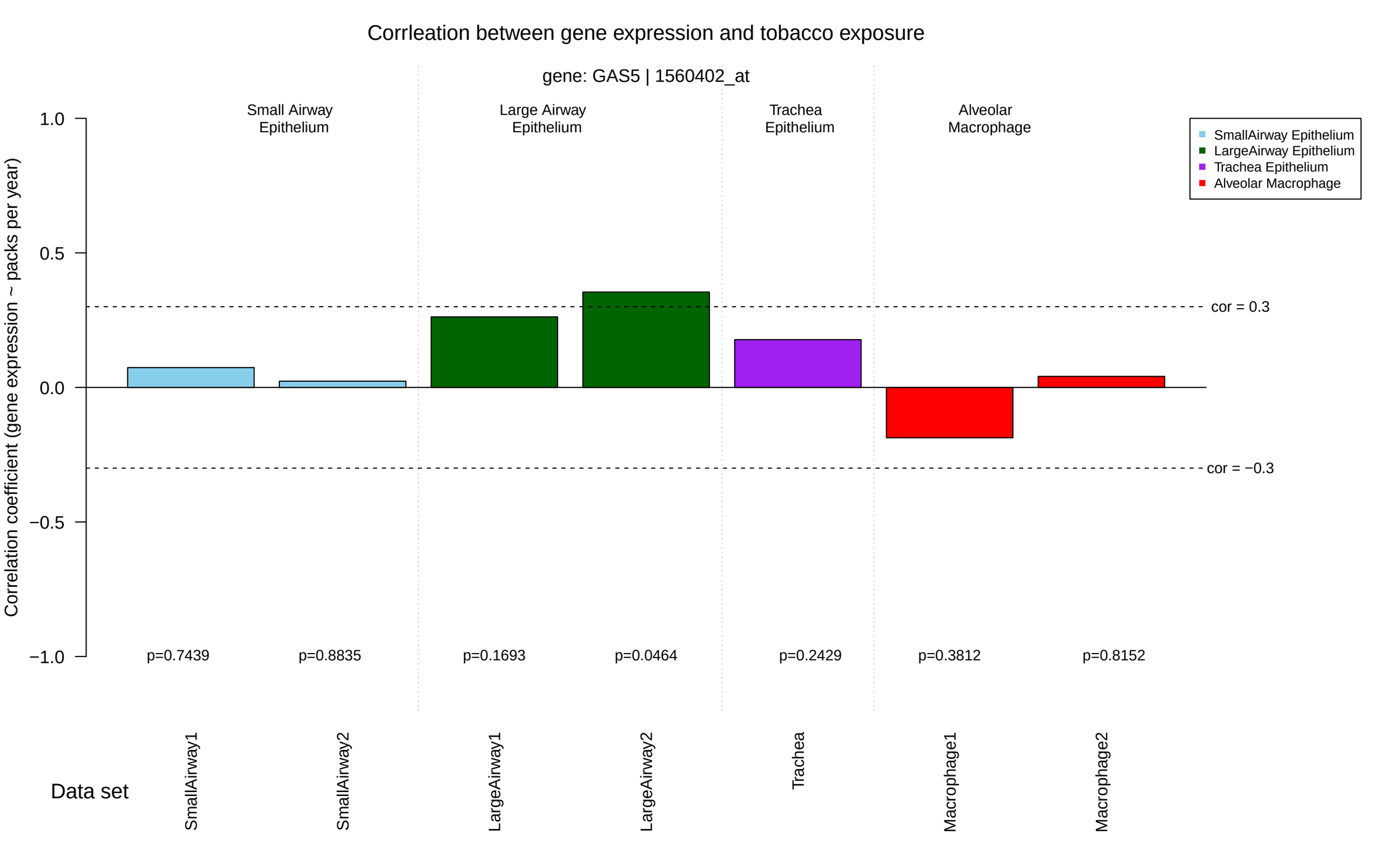

set: LargeAirway1 (p=0.4242)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4769)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6801)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3213)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8728)
GAS5 | 1560402_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAS5 | 1560402_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7439
p=0.8835
p=0.1693
p=0.0464
p=0.2429
p=0.3812
p=0.8152
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2